

Assessment of a Community Pharmacy-Based Program for Patients with Asthma

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Study Objective. To implement and assess a community-based pharmaceutical care program for patients with asthma.

Design. Prospective, randomized, controlled trial.

Setting. Community pharmacies (11 control, 11 intervention) in Malta.

Patients. Community-dwelling patients with asthma.

Interventions. A comprehensive asthma education and monitoring program was implemented. Intervention patients received verbal counseling, an educational video, an information leaflet, and subsequent monitoring with reinforcement; control patients received routine dispensing services.

Measurements and Main Results. Parameters assessed at baseline and at 4, 8, and 12 months were health-related quality of life, peak expiratory flow (PEF), inhaler technique, compliance with therapy, hospitalization rates, days lost from work, asthma symptoms, and patient satisfaction. Health-related quality of life of the intervention patients improved at 12 months ($p=0.044$). In the same time period, PEF significantly decreased in control patients compared with intervention patients ($p=0.009$) whereas inhaler technique improved in the intervention group ($p=0.021$). There were significantly fewer self-reported hospitalizations in intervention patients.

Conclusions. A community-based pharmaceutical care program was appreciated by the participants and had a positive impact on the vitality of patients with asthma, inhaler technique, and PEF.

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The delivery of care to patients with asthma is a complex process that could be improved by using a team of health care professionals with a variety of skills.¹ The pharmacist's contribution to patient care through education and monitoring, and in particular through sharing responsibility for drug therapy, is an approach being advocated to optimize drug therapy and improve patients' quality of life.² Various studies in the United States and the United Kingdom

have reported an improvement in various aspects of asthma management after intervention by a pharmacist.^{3–5} In many other countries, the involvement of pharmacists in this aspect of patient care is minimal.

Pharmaceutical services in Malta may be broadly classified into two groups: private community pharmacies and government pharmaceutical services. Government pharmaceutical services include seven dispensaries that service different regions of the island. They dispense drugs free of charge to certain categories of patients, including those with chronic diseases such as asthma. In addition to these dispensaries, a postal system for obtaining drugs is available through small government clinics located in nearly every town and village. Only basic dispensing is performed within these clinics, as

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the service is mainly one of supply and distribution. At the government dispensaries, pharmacists and technicians dispense drugs, whereas patients who collect their drugs through the postal system come in contact with nurses who hand out prefilled prescriptions. Therefore, within the government health service, contact between pharmacists and patients is limited. Patients entitled to receive free drugs may collect their prescriptions only through the government pharmaceutical services.

Adult and adolescent patients older than 14 years in whom asthma is diagnosed or suspected usually are referred to the outpatient asthma clinic at the main general hospital on the island, where they are placed in the care of a specialist in respiratory medicine. However, patients do not receive any formal education about their disease state or their drug therapy. Self-monitoring of peak expiratory flow (PEF) is limited, due to the lack of availability of both peak flow meters and diary cards. As a matter of policy, patients younger than 14 years are referred to pediatric outpatient clinics.

Malta's 198 independent, privately-owned community pharmacies are highly accessible, with a population:pharmacy ratio of approximately 1900:1. These pharmacies are run strictly on a profit basis and are not subsidized by the state in any way. Pharmacists practicing in community pharmacies are more patient oriented than those practicing in the state service. The present system of health care, however, excludes the community pharmacist from managing asthma and other chronic diseases because these are managed through the state system and secondary care.

The objective of our controlled clinical trial was to examine the impact of a community pharmacy-based education and monitoring program for patients with asthma on a range of patient-specific asthma management outcomes. To facilitate the trial, a system was established to encourage cooperation between government and private services, including the organization of a distribution system for free asthma drug therapy through community pharmacies to eligible patients.

Methods

This was a randomized, controlled, longitudinal, prospective trial. Ethical approval to perform the study was obtained from the Ethical Committee, Faculty of Medicine, the University of Malta.

Special permission was granted from health authorities to decentralize the dispensing of government drugs through participating community pharmacies. On recruitment, a record of the drug entitlement for each patient was documented. Based on this information, a drug order was placed at the Government Medical Stores each month. The study coordinator collected these drugs, subdivided them into stock for each pharmacy, and delivered them to the study sites once/month.

Pharmacy Recruitment and Pharmacist Training

To facilitate patient recruitment, we sought collaboration with specialists in respiratory medicine at the main asthma outpatient clinic on the island. Using a list of private community pharmacies provided by the Department of Health, 22 pharmacies were selected from the 11 geographic areas (2 from each area) with the highest number of patients with asthma registered at the clinic. These pharmacies were invited to take part in the study. Within each geographic area, sites were randomly assigned as either control or intervention. The managing pharmacist at each site was invited to participate in the study and was informed of the allocation of control or intervention status.

A manual was prepared in the form of a self-study program. The manual contained two sections, the first of which dealt with the pathophysiology of asthma and its treatment, including standard intervention instructions. The second section provided details of the outcome measures and data collection instruments to be used in the study. The most recent guidelines of the British Thoracic Society available at that time (1993) were used as a basis for the therapeutic management of asthma.⁶ Intervention pharmacists received both sections of the manual, whereas control pharmacists received section two only.

Pharmacist training took place over 2 evenings. The manual was sent to each intervention pharmacist 1 month before the training sessions. The first evening, which was open only to intervention pharmacists, reviewed the pathology and therapeutics related to asthma and discussed the education-structured pharmaceutical care of asthmatic patients. During this first evening, which was conducted by the principal investigator with one of the participating respiratory specialists, emphasis was placed on patient education, patient monitoring, and

treatment regimens. Intervention pharmacists were shown samples from 5-minute education videotapes (prepared in-house) on the optimal use of the inhaler devices. Intervention pharmacists' knowledge was assessed informally through an interactive workshop using case studies and problems to ensure that they were competent to deliver the care model. The second evening, which all participating pharmacists attended, dealt with the evaluation instruments and how they were to be used during the research program. Thus, control pharmacists were instructed only on the use of evaluation instruments, including when and how these instruments were to be used; they were not instructed on the pharmaceutical care program.

Patient Recruitment

Patients registered at the asthma clinic were invited to participate in the study through recruitment sessions. During these sessions, patients who gave written informed consent to participate in the study were informed of two pharmacies participating in the research project within their area. Exclusion criteria were presence of other significant pulmonary disease (e.g., pulmonary fibrosis, emphysema, bronchitis, carcinoma), presence of any condition that would hinder performance of pulmonary function tests or completion of questionnaires (e.g., mental retardation), and age below 14 years.

Pharmaceutical Care Interventions

The pharmaceutical care interventions implemented by intervention pharmacies focused on two issues: patient education and patient monitoring. Education addressed asthma pathology, including avoidance of triggering factors and use of inhaled drugs and peak flow meters. Verbal education and demonstration of inhaler technique were supported by written information and provision of a short videotape for home viewing. Patients were monitored by supplying them with a peak flow meter and asking them to record their PEF rate in the morning and evening, together with asthma symptoms, on a diary card. Patients were instructed to present their diary card to their community pharmacist for review monthly when the patients collected their asthma drugs. They were encouraged to use spacers, but these had to be purchased by the patients.

Intervention pharmacists were provided with a patient profile that included information on the

patient's best peak flow value, smoking history, other disease states, known drug allergies, and prescribed drugs. These data were retrieved from each patient's asthma clinic file. When the patient returned to the pharmacy each month, the pharmacist inquired about asthma symptoms and problems encountered with treatment. The pharmacist also reviewed the patient's PEF record, compared it with the patient's personal best, and advised the patient as necessary. When required, the pharmacist reviewed the patient's inhaler and PEF technique, and reviewed the patient's asthma status and drug therapy.

This information was recorded in the patient profile, together with any referral to the asthma clinic, suggested treatment changes that were relayed to the patient's physician, and time spent with the patient. Patients attending control pharmacies were given their prescribed drugs and informed of the dosage regimen but received no other assistance.

Patients were registered with the pharmacy of their choice and were unable to switch between control and intervention pharmacies, as they could obtain their drugs only from the pharmacy where they were registered.

Baseline and Outcome Measures

Baseline and outcome measures were performed by the study coordinator. Four parameters were assessed at baseline and at 4, 8, and 12 months. The first was the patient's health-related quality of life as assessed by the Short Form (SF)-36, a generic instrument, and assessment by disease-specific instruments was done with the Living with Asthma Questionnaire (LWAQ) for adults or the Childhood Asthma Questionnaire (CAQ) for those aged 14–17 years.^{7–9} The SF-36 generates a score ranging from 0 (worst possible health) to 100 (best possible health) for each of the following eight multi-item dimensions: physical functioning, role limitations due to physical problems, role limitations due to emotional problems, social functioning, mental health, energy and vitality, pain, and general health perceptions. The LWAQ is a 68-item, 11-domain unidimensional asthma-specific quality-of-life questionnaire that generates an overall score on a scale of 0–2, with 0 indicating excellent quality of life and 2 indicating very poor quality of life.

The second parameter assessed by pharmacists was the patient's PEF rate, determined with a peak flow meter. Third, pharmacists used a

Table 1. Demographic Comparison of Intervention and Control Patients

	Intervention Group No. (%)	Control Group No. (%)	Total No. (%)
Men ^a	37 (42.0)	40 (60.6)	77 (50.6)
Women	49 (57.0)	26 (39.4)	75 (49.4)
Age, yrs			
14–17	8 (9.3)	5 (7.6)	13 (8.5)
18–44	36 (41.9)	18 (27.3)	54 (35.5)
45–64	36 (41.9)	28 (42.4)	64 (42.1)
> 65	6 (7.0)	15 (22.7)	21 (13.8)
Mean age	41.3 ± 18.35 ^b	45.88 ± 18.11	43.2 ± 18.32
Age range	14–82	14–79	14–82

^aStatistically significant difference, $p < 0.05$, Fisher exact test.^bStatistically significant difference, $p < 0.05$, χ^2 test.

categorical scoring system⁵ to assess the patient's inhaler technique. The patient was observed while using the inhaler and a score (1 = correct, 0 = incorrect) was given for each step according to standard instructions. An overall score was obtained by adding the scores for each step and then converting the total to a percentage. When a patient omitted a critical step, such as forgetting to remove the mouthpiece cover from a metered-dose inhaler or forgetting to load a dry powder inhaler, the pharmacist assigned a score of 0.

Finally, pharmacists assessed each patient's asthma control using an outcome measures questionnaire that involved a structured patient interview. Patients were asked how often they forgot to take their prescribed drugs, how often they were admitted to the hospital, and if hospitalized, the length of their hospitalization. Patients also were asked about days absent from work and visits to the doctor.

Patients' subjective opinions of the services provided were obtained through a structured patient satisfaction questionnaire, which was administered at baseline and 12 months only. The patients were asked to rank, on Likert-type scales (excellent to poor), the care, information, and advice they received. They also were asked to state their agreement (strongly agree to strongly disagree) with a range of statements on the ease of interaction with the pharmacist and the pharmacist's knowledge and ability to counsel them about asthma and its treatment.

Data Analysis

Data obtained at baseline and the assessment points from quality-of-life instruments, the

Table 2. Number of Patients at Baseline and at Each Assessment Period

Assessment Period	No. of Patients		
	Intervention	Control	Total
Baseline	86	66	152
4 months	64	58	122
8 months	61	57	118
12 months	64	55	119

outcome measures questionnaire, PEF rate measurements, and inhaler technique assessments were analyzed using repeated-measures analysis of variance (ANOVA). In addition, changes between baseline and 12 months for the various measures were compared within and between the two groups of patients using *t* tests, the Wilcoxon signed rank test (within-group analysis), and the Mann-Whitney U test (between-group analysis), as appropriate. A *p* value below 0.05 was considered statistically significant.

Results

A total of 152 patients were recruited to participate in the study: 86 in the intervention group and 66 in the control group. Table 1 provides relevant demographic information. The number of men in the intervention group was significantly lower than in the control group ($p = 0.035$; Fisher exact test). The intervention group also had a significantly younger mean age compared with the control group (χ^2 test; $p = 0.03$). Nearly 20% of the patients dropped out of the study within the first 4 months; most of these (73.3%) were in the intervention group. Table 2 summarizes the number of patients at baseline and at each of the assessment periods.

Health-Related Quality of Life

Analysis of SF-36 data showed no significant differences between the two groups over time for the eight dimensions. However, there was a significant difference between the two groups for the change in score in the vitality dimension between baseline and 12 months ($p=0.001$; ANOVA with age and gender as covariates). The mean change $\pm$ SD was positive for the intervention group (4.29 ± 20.9) and negative for the control group (-8.87 ± 21.8).

In terms of dropouts from the study, analysis of baseline scores for the SF-36 dimensions showed no significant differences between those control patients who dropped out and their counterparts who remained until the 12-month assessment ($p>0.05$). In the case of the intervention patients, the same pattern was observed, with the exceptions of physical functioning, vitality, and bodily pain. In these dimensions, the scores were significantly higher at baseline (i.e., suggesting these patients were more healthy) in patients absent at 12 months compared with those present at 12 months.

When comparing LWAQ scores in the control versus intervention groups at each of the assessment periods, intervention patients reported lower scores (i.e., better quality of life) than patients in the control group at 4, 8, and 12 months. However, using repeated-measures ANOVA with gender and age as covariates, LWAQ scores were not significantly different between the two groups. When analyzing the data for each group over time, it is evident that for the 60 patients in the intervention group who completed the study, the mean score at 12 months (0.96 ± 0.38) was significantly lower than at baseline (1.03 ± 0.36 , $p=0.044$; paired t test), indicating an improvement in quality of life over the duration of the study. The corresponding comparison for the control group was not statistically significant. When considering the change in a patient's score between baseline and 12 months (12-month score minus baseline score), the mean change for the intervention group (-0.0064 ± 0.24) was borderline significantly different from the mean change for the control group ($+0.0061 \pm 0.35$, $p=0.053$; ANOVA with gender and age as covariates). The differences were, however, very small and are unlikely to be of clinical significance.

As only 13 patients aged 14–17 years were recruited, no meaningful data on health-related

quality of life in this age group using the CAQ were collected.

Assessment of PEF

The results of the PEF rate demonstrated that at baseline, the mean PEF of the control patients was higher than that of the intervention group (390.12 ± 137.91 L/min vs 374.93 ± 134.08 L/min). However, at 4, 8, and 12 months, the mean PEF values for the control patients were lower than those of the intervention patients. Using repeated-measures ANOVA with gender and age as covariates, PEF was not significantly different in the two groups ($p>0.05$).

For the 55 patients in the control group with PEF measurements at baseline and 12 months, the mean PEF at baseline (377 ± 131 L/min) was significantly higher than the mean PEF at 12 months (340 ± 115 L/min, $p=0.009$; paired t test). The corresponding comparison for the intervention group was nonsignificant. The change (by subtraction) between a patient's 12-month PEF and baseline PEF was not significantly different between the two groups (ANOVA with gender and age as covariates).

Assessment of Inhaler Technique

The inhaler scores resulting from the categorical scoring system were calculated as percentages to eliminate differences between inhaler types. This resulted in scores of 0–100%. This categorical scoring system yielded percentage scores of 0%, 75%, 83%, 87.5%, and 100%. For analysis, a classification scheme was used. Those scoring 0% were placed in class 1, whereas those scoring 75% or greater were placed in class 2.

At baseline, both groups had similar inhaler scores, and a relatively low percentage of patients scored 75% or greater (22.1% in the intervention group and 22.7% in the control group). Throughout the study, patients in both groups showed an increasing trend to score above 75%. For the assessments at 4, 8, and 12 months, the percentage of intervention patients scoring above 75% was 39.5%, 44.2%, and 58.1%, respectively. Corresponding scores for the control group were 28.85%, 48.5%, and 47.0%, respectively. Table 3 summarizes the number of patients who achieved changes in scores between baseline and 12 months. The difference between the 12-month score and the baseline score was significantly higher in the intervention group than in the control group ($p=0.021$; Mann-Whitney U test).

Table 3. Number of Patients Who Achieved Changes in Inhaler Scores between Baseline and 12-Month Assessment Periods

Change	No. of Patients	
	Control	Intervention
No change	26	24
Decrease	5	0
Increase	24 ^a	40 ^a

^a $p < 0.001$ (McNemar's test) when comparing baseline with 12-month scores.

Self-Reported Compliance with Inhalers

When asked how often they forgot to use their inhalers, most patients stated that they forgot about 4 times/month. This response was fairly constant over time for both the intervention and control patients (averaging 65% in the intervention group and 60% in the control group). An average of 17% in the intervention group and 27% in the control group admitted to forgetting once/day, whereas the remainder stated that they forgot to use their inhalers about 2–3 times/week. There were no significant differences between the groups at 12 months ($p > 0.05$; χ^2 test).

Self-Reported Hospitalization Rates, Doctor Visits, and Days Lost from Work or Other Activities

A minority of participants were hospitalized due to asthma exacerbations during the analysis period. No members of the intervention group were hospitalized, but eight patients in the control group were hospitalized due to exacerbations of asthma. These patients had a mean length of hospital stay of 3.57 ± 2.07 days. The self-reported hospitalization rates were significantly different between the two groups ($p = 0.002$; Fisher exact test). On average, 49% of the control patients and 48% of the intervention patients said that they did not visit their physicians in relation to their asthma, whereas an average of 40% in the control group and 39% in the intervention group said that they visited the physician only once regarding their asthma. There were no significant differences between the groups ($p > 0.05$; χ^2 test). The question regarding days lost from work was not applicable to 44.7% of participants, as 33.5% were housewives and 11.2% were pensioners. On average, patients in both groups missed 2–5 days of work or school. Statistical analysis did not reveal any significant differences in the responses regarding days lost

from work or school between the groups ($p > 0.05$; χ^2 test).

Self-Reported Symptoms

Patients were asked about the severity of their wheezing, cough, chest tightness, and shortness of breath. Over 70% of patients in both the control and intervention groups reported that they experienced these symptoms some of the time or none of the time at baseline and at each of the assessment periods. Up to 30% reported feeling these symptoms all the time or most of the time. At 12 months, significantly more patients in the intervention group than in the control group (80% vs 64%) reported no wheezing or only nighttime wheezing ($p = 0.051$; χ^2 test). A significantly smaller percentage of the intervention group than the control group (20% vs 36%) reported nighttime wheezing all the time or most of the time at the final assessment point.

Assessment of Patient Satisfaction with Services Provided

Data regarding patient satisfaction with the services provided by pharmacists were collected by means of an interviewer-administered questionnaire. At baseline, patients were receiving their drugs from a government dispensary; therefore, responses for this assessment point refer to government pharmacists. However, at 12 months, patients' responses were based on the services provided by the private community pharmacists in either control or intervention pharmacies.

At baseline, both control and intervention patients classified the overall service provided as poor. However, after the 12-month study period, the responses obtained from both groups of patients were very positive. At baseline, none of the patients reported receiving excellent or very good care from the pharmacist; however, at 12 months, 51% of control patients and 64% of intervention patients felt that this was the case. When assessing the information and advice provided by the pharmacist at the 12-month assessment period, control patients (81.8%) were more likely to rate this service as poor than intervention patients (20.3%, $p < 0.0001$; χ^2 test). At baseline nearly all patients declared that they never discussed their asthma or their prescribed drugs with the pharmacist. However, at the end of the study, 22% of the intervention group had discussed their condition about 1–3 times; 33% recalled 4–6 discussions; 25% cited 6–12

discussions; and 2% stated that they had participated in more than 12 discussions with the pharmacist over the 1-year period. On average, the pharmacists reported spending 5–10 minutes/session discussing asthma with the intervention patients. Most control patients (80%) still stated that they never spoke to the pharmacist about their asthma or drugs, whereas 20% of control patients reported some contact with their pharmacist regarding their condition and its management.

The professional relationship with the pharmacist improved in both groups of patients, but especially so in the intervention group. At 12 months, intervention patients were more likely to find it easy to approach their pharmacist (64% intervention vs 29% control, $p < 0.0001$; χ^2 test) and were more likely to ask the pharmacist questions about their drugs (70% intervention vs 43% control, $p = 0.02$; χ^2 test).

Discussion

This study demonstrated that pharmaceutical care provision was well received by patients, despite the high level of participant dropout. The baseline SF-36 scores of those who dropped out were compared with the baseline scores of those who remained in the study until 12 months. There were no differences in any dimensions for the control patients, but intervention patients who dropped out achieved higher scores in three dimensions at baseline compared with baseline scores for those who completed the study. This suggests that the patients who dropped out were more healthy at baseline than their counterparts. The patients who dropped out of the study were contacted by phone and asked about their reason for discontinuing the study. The main reason given was that the self-monitoring part of the program was inconvenient. Other dropouts said that they had no time to collect their drugs in person and therefore could not be monitored by the pharmacist; a few were concerned that a fee would be introduced for drugs because these were being administered by a private pharmacy. This would suggest that patients dropped out for logistical reasons and not because they were more sick than those who remained in the study. Furthermore, this method of patient care was markedly different from that normally experienced by patients in terms of pharmacy input.

The SF-36 results indicate that the pharmacists'

intervention had an effect on patients' quality of life. However, as expected, the improvement in quality of life was small, since these patients were already closely monitored by respiratory consultants. In contrast, a similar randomized, controlled trial, in which the intervention was carried out by specialized asthma nurses in 41 general practices in the London area, did not detect any improvement in asthma-related quality of life or in the quality of asthma care.¹¹ This lack of effect was attributed to the large number of patients who required management in primary care and the high turnover of practice nurses in inner city areas.

The value of educating patients in the use of inhaler technique is evident from our results and concurs with other studies.^{3, 5} Our study also detected an improvement in the control group, which could be attributed to initiation of inhaler technique instruction by one of the participating hospital physicians during the study. However, the higher scores achieved by the intervention patients suggest that the combination of methods employed by the pharmacists produced better results, thus reinforcing the benefits of using such methods.^{12–14} Data relating to compliance with inhalers did not differentiate between the control and intervention groups, possibly due to the method of self-reported data employed. However, other methods, such as examination of refill records, would have failed to discriminate because patients in Malta who are entitled to receive free drugs collect their drugs regularly irrespective of whether they need them. Hoarding of free drugs is a well-known phenomenon in Malta.

Patients in the intervention group reported high levels of satisfaction with the service and care provided by their pharmacist. The results suggest that the pharmacists providing pharmaceutical care were viewed as being more approachable by intervention patients than by control patients. Patients in the intervention group were also more likely to perceive the pharmacist as a health care professional. Similarly, the Grampian Asthma Study of Integrated Care (GRASSIC) found that patient and physician confidence was fostered by the experience of integrated care.¹⁵ Asking patients to report on the quality of their care may help bring the patient and the health care professional closer, thus providing a more holistic view of the patient, rather than just a clinical picture.¹⁶

The results of this study suggest that pharmacists need to become more proactive in

terms of integrating themselves more fully into chronic disease management. Clearly, the extent of interchange between Maltese intervention pharmacists and their patients could have been improved. A recent paper highlighted that in community pharmacies in the northeast of Scotland, asthma counseling was infrequent.¹⁷ This was attributed to limited facilities within pharmacies, lack of pharmacist knowledge of some aspects of asthma management, and patient attitudes (i.e., lack of awareness that pharmacists can fulfill this role).

The importance of training and support for pharmacists involved in asthma care was highlighted by a recent pilot study conducted in southern England.¹⁸ In addition, research in the U.S. has demonstrated the importance of patient involvement and provided support for new modes of service delivery. One such study explored the feasibility of a community asthma management network program in community pharmacies in West Virginia.¹⁹ The study sought to assess the impact of a community-based asthma management program aimed at empowering patients in such areas as asthma self-management, knowledge, drug compliance, and quality of life. Preliminary findings suggest that the program is successful due to appropriate asthma education delivery by pharmacists.

Summary

Pharmaceutical care offers a challenge to pharmacists to extend their involvement in patient-centered activities. Our study represented a major change in conventional practice for Maltese pharmacists. It demonstrates that both the profession and patient groups must accept that pharmaceutical intervention in chronic disease management can be a major step forward. The study showed that a community-based pharmaceutical care program was appreciated by the participants and had a positive impact on patients' vitality, inhaler technique, and PEF. The self-reported decrease in hospitalizations suggests that the intervention may have positively affected asthma management in these patients, perhaps with economic benefits. A more definitive assessment of these benefits in a larger study, however, is required.²⁰

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